B. TECH-ME-III-II Sem

23A03604c	CONTROL SYSTEMS (Professional Elective-II)	L	T	P	C
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Course objectives: The objectives of the course are to	
1	Fundamentals of control systems, mathematical modeling, and transfer function derivation for electro-mechanical components.
2	Knowledge on system representation, and to introduce classical control design techniques.
3	To equip students with the ability to represent and simplify control, and to design effective controllers for achieving desired system performance.
4	Knowledge and skills to analyze control system stability and performance in the frequency domain, using Bode, polar, and transfer functions.
5	State-space analysis of different models from block diagrams, and concepts of controllability, observability, and state transition matrices.

COURSE OUTCOMES On successful completion of this course the student will be able to		
1	Model dynamic systems, distinguish between loop controls, analyze feedback effects, and derive transfer functions for DC/AC servo motors and synchro devices.	*L1,L2,L4*
2	Analyze and simplify control systems, and create effective controllers such as lag, lead, lead-lag, and PID to meet system performance specifications.	L2,L4,L6
3	Analyze and simplify control systems using block diagrams and signal flow graphs, and design appropriate controllers to achieve desired system performance.	L3,L4,L6
4	Analyze the frequency response of control systems using Bode, polar, and evaluate system stability and performance such as gain margin and phase margin.	L1,L4,L5
5	Develop state-space models from block diagrams, analyze system controllability and observability, and solve time-invariant state equations using the state transition matrix.	L2,L4,L6

UNIT - I

BASICS IN CONTROL SYSTEM AND TRANSFER FUNCTION: Introduction of Control Systems, Various types of systems (Open Loop and closed loop) and their differences- Classification and Feed-Back Characteristics of control system- Effects of feedback. Mathematical models – Differential equations, Translational and Rotational mechanical systems. Transfer Function of DC Servo motor - AC Servo motor- Synchro transmitter and Receiver.

UNIT - II

REPRESENTATION OF TRANSFER FUNCTION AND CONTROL DESIGN TECHNIQUES: Block diagram representation of systems considering electrical systems as examples. Block diagram algebra – Representation by Signal flow graph - Reduction using Mason's gain formula. Compensation techniques – Lag, Lead, Lead-Lag Controllers design, PID Controllers.

UNIT - III

TIME RESPONSE ANALYSIS: Standard test signals - Time response of first order systems – Characteristic Equation of Feedback control systems, Transient response of second order systems - Time domain specifications – Steady state response - Steady state errors and error constants – Effects of proportional derivative, proportional integral systems. STABILITY ANALYSIS: The concept of stability – Routh's stability criterion – qualitative stability and conditional stability – limitations of Routh's stability. The root locus concept - construction of root locieffects of adding poles and zeros to G(s) H(s) on the root loci.

UNIT - IV

FREQUENCY RESPONSE ANALYSIS: Introduction, Frequency domain Specifications-Bode diagrams Determination of Frequency domain specifications and transfer function from the Bode Diagram-Phase margin and Gain margin Stability Analysis from Bode Plots. **STABILITY ANALYSIS IN FREQUENCY DOMAIN**: Polar Plots-Nyquist Plots-Stability Analysis.

UNIT - V

STATE SPACE ANALYSIS: Concepts of state, state variables and state model, derivation of state models from block diagrams, Diagonalization- Solving the Time invariant state Equations- State Transition Matrix and it's Properties – Concepts of Controllability and Observability.

TEXT BOOKS:

1. I. J. Nagrath, M .Gopal (2011), Control Systems Engineering, 5th edition, New Age International (P) Limited, New Delhi, India.
2. Benjamin. C. Kuo (2003), Automatic Control Systems, 8th edition, John Wiley and Son's, USA.

REFERENCE BOOKS:

1. K. Ogata (2008), Modern Control Engineering, 4th edition, Prentice Hall of India Pvt. Ltd, New Delhi. 2. N. K. Sinha (2008), Control System.
2. Prof. Vishwajit K. Barbudhe Control system Engineering National Press (2020)
3. Richard Dorf and Robert Bishop Modern Control Systems Pear son, 13th edition (2016)

Online Learning Resources:

1. https://nptel.ac.in/courses/107106081
2. https://nptel.ac.in/courses/108107115
3. https://nptel.ac.in/courses/108103007
4. https://nptel.ac.in/courses/115108104